

Eigen Values and Eigen Vectors of Matrices

Eigenvalues and Eigenvectors: Let $\mathcal{A}$ be an n -square matrix. A scalar λ is called an eigenvalue or characteristic root of $\mathcal{A}$ if there exists a non-zero column vector v such that,

$$Av = \lambda v \quad \text{or} \quad (A - \lambda I)v = 0$$

Where, I is the n -square identity matrix. Every vector satisfying this relation is called an eigenvector or characteristic vector of $\mathcal{A}$ associated with the eigenvalue λ .

Spectrum: The set of all eigenvalues of a matrix $\mathcal{A}$ is called the spectrum of $\mathcal{A}$.

Eigenspace: If $\mathcal{A}$ be an n -square matrix and λ be an eigenvalue of $\mathcal{A}$, then the set of all vectors satisfying the relation $Av = \lambda v$ including the zero vector is called an eigenspace of $\mathcal{A}$ corresponding to λ .

Characteristic Matrix: If $\mathcal{A}$ be an n -square matrix and λ be an eigenvalue of $\mathcal{A}$, then the matrix $A - \lambda I_n$ or $\lambda I_n - A$ is called a characteristic matrix of $\mathcal{A}$.

Characteristic Polynomial: If $\mathcal{A}$ be an n -square matrix and λ be an eigenvalue of $\mathcal{A}$, then the determinant of the characteristic matrix $A - \lambda I_n$ or $\lambda I_n - A$ is called a characteristic polynomial of $\mathcal{A}$.

$$\text{i.e., } \Delta = |A - \lambda I_n|$$

$$\text{or, } \Delta = |\lambda I_n - A|.$$

Characteristic Equation: If $\mathcal{A}$ be an n -square matrix and λ be an eigenvalue of $\mathcal{A}$, then the equation $|A - \lambda I_n| = 0$ or $|\lambda I_n - A| = 0$ is called a characteristic equation of $\mathcal{A}$.

NOTE: If $\mathcal{A}$ be an $n \times n$ matrix then,

1. An eigenvalue of $\mathcal{A}$ is a scalar such that $|A - \lambda I_n| = 0$
2. An eigenvectors of $\mathcal{A}$ corresponding to λ are the non-zero solutions of $(A - \lambda I_n)v = 0$.

Algebraic Multiplicity: The number of times an eigenvalue occurs is called its algebraic multiplicity. For example, if $\lambda = -2, -2, 0, 3, 5$; then algebraic multiplicities of $-2, 0, 3, 5$ are 2, 1, 2, 1 respectively.

Geometric Multiplicity: The geometric multiplicity of an eigenvalue is the dimension of the eigenspace associated with that eigenvalue.

NOTE: The geometric multiplicity of an eigenvalue is either less than or, equal to the algebraic multiplicity of that eigenvalue.

NOTE: If $\mathcal{A}$ is an $n \times n$ triangular matrix (upper triangular, lower triangular, or diagonal), then the eigenvalues of $\mathcal{A}$ are the entries on the main diagonal of $\mathcal{A}$.

Example: If $A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ 0 & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & a_{nn} \end{pmatrix}$, then its eigenvalues are, $a_{11}, a_{22}, \dots, a_{nn}$.

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NOTE:

- 1). Every polynomial equation of odd degree with real coefficients does possess a real solution.
- 2). $\lambda=0$ is an eigenvalue of a matrix A iff A is singular.
- 3). All eigenvalues of a matrix A are non-zero iff A is non-singular.
- 4). If a matrix A has an eigenvalue λ and $k \in \mathbb{N}$, then λ^k will become an eigenvalue of A^k .
- 5). Same eigenvalues belong to similar matrices.
- 6). Characteristic values of a symmetric matrix are real numbers.
- 7). Characteristic values of a skew-symmetric matrix are either zeroes or imaginary numbers.
- 8). Characteristic values of a Hermitian matrix are real numbers.
- 9). Characteristic values of an orthogonal matrix are 1 or -1.
- 10). The eigenvalues of a matrix A are same as the eigenvalues of A^t .
- 11). For, a non-singular matrix, the inverse of an eigenvalue will be the eigenvalue of the inverse matrix.

Cayley-Hamilton Theorem: Every square matrix satisfies its characteristic equation.

Problem-01: Verify Cayley-Hamilton Theorem for $A = \begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix}$.

Solution: The given matrix is,

$$A = \begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix}$$

The characteristic matrix of A is,

$$\begin{aligned} A - \lambda I &= \begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} 1-\lambda & 3 & 7 \\ 4 & 2-\lambda & 3 \\ 1 & 2 & 1-\lambda \end{pmatrix} \end{aligned}$$

The characteristic polynomial of A is,

$$\Delta = |A - \lambda I|$$

$$\begin{aligned}
 &= \begin{vmatrix} 1-\lambda & 3 & 7 \\ 4 & 2-\lambda & 3 \\ 1 & 2 & 1-\lambda \end{vmatrix} \\
 &= (1-\lambda)\{(2-\lambda)(1-\lambda)-6\}-3\{4(1-\lambda)-3\}+7\{8-(2-\lambda)\} \\
 &= (1-\lambda)(\lambda^2-3\lambda-4)-3(1-4\lambda)+7(6+\lambda) \\
 &= -\lambda^3+4\lambda^2+\lambda-4-3+12\lambda+42+7\lambda \\
 &= -\lambda^3+4\lambda^2+20\lambda+35
 \end{aligned}$$

The characteristic equation of $\mathcal{A}$ is,

$$|A - \lambda I| = 0$$

$$\Rightarrow -\lambda^3 + 4\lambda^2 + 20\lambda + 35 = 0$$

$$\Rightarrow \lambda^3 - 4\lambda^2 - 20\lambda - 35 = 0$$

To verify Cayley-Hamilton Theorem we need to show:

$$A^3 - 4A^2 - 20A - 35I = 0$$

Now we have,

$$\begin{aligned}
 A^2 &= \begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 1+12+7 & 3+6+14 & 7+9+7 \\ 4+8+3 & 12+4+6 & 28+6+3 \\ 1+8+1 & 3+4+2 & 7+6+1 \end{pmatrix} \\
 &= \begin{pmatrix} 20 & 23 & 23 \\ 15 & 22 & 37 \\ 10 & 9 & 14 \end{pmatrix}
 \end{aligned}$$

$$\begin{aligned}
 A^3 &= AA^2 \\
 &= \begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 20 & 23 & 23 \\ 15 & 22 & 37 \\ 10 & 9 & 14 \end{pmatrix} \\
 &= \begin{pmatrix} 135 & 152 & 232 \\ 140 & 163 & 208 \\ 60 & 76 & 111 \end{pmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \therefore A^3 - 4A^2 - 20A - 35I &= \begin{pmatrix} 135 & 152 & 232 \\ 140 & 163 & 208 \\ 60 & 76 & 111 \end{pmatrix} - 4 \begin{pmatrix} 20 & 23 & 23 \\ 15 & 22 & 37 \\ 10 & 9 & 14 \end{pmatrix} - 20 \begin{pmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{pmatrix} - 35 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 135 & 152 & 232 \\ 140 & 163 & 208 \\ 60 & 76 & 111 \end{pmatrix} - \begin{pmatrix} 80 & 92 & 92 \\ 60 & 88 & 148 \\ 40 & 36 & 56 \end{pmatrix} - \begin{pmatrix} 20 & 60 & 140 \\ 80 & 40 & 60 \\ 20 & 40 & 20 \end{pmatrix} - \begin{pmatrix} 35 & 0 & 0 \\ 0 & 35 & 0 \\ 0 & 0 & 35 \end{pmatrix}
 \end{aligned}$$

$$= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Hence, the Cayley-Hamilton Theorem is verified for the given matrix.

Problem-02: Find the inverse of the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix}$ by using Cayley-Hamilton

Theorem.

Solution: The given matrix is,

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix}$$

The characteristic matrix of A is,

$$\begin{aligned} A - \lambda I &= \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} 1-\lambda & 2 & 3 \\ 2 & -1-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{pmatrix} \end{aligned}$$

The characteristic polynomial of A is,

$$\begin{aligned} \Delta &= |A - \lambda I| \\ &= \begin{vmatrix} 1-\lambda & 2 & 3 \\ 2 & -1-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{vmatrix} \\ &= (1-\lambda)\{(-1-\lambda)(1-\lambda)-1\} - 2\{2(1-\lambda)-3\} + 3\{2-(-1-\lambda)\} \\ &= (1-\lambda)(\lambda^2 - 2) - 2(-1-2\lambda) + 3(3+\lambda) \\ &= -\lambda^3 + \lambda^2 + 2\lambda + 2 + 4\lambda + 9 + 3\lambda \\ &= -\lambda^3 + \lambda^2 + 9\lambda + 11 \end{aligned}$$

The characteristic equation of A is,

$$|A - \lambda I| = 0$$

$$\Rightarrow -\lambda^3 + \lambda^2 + 9\lambda + 11 = 0$$

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$$\Rightarrow \lambda^3 - \lambda^2 - 9\lambda - 11 = 0$$

By Cayley-Hamilton Theorem we can write:

$$A^3 - A^2 - 9A - 11I = 0$$

Multiplying both sides by A^{-1} we have,

$$A^2 - A - 9I - 11A^{-1} = 0$$

$$\Rightarrow A^{-1} = \frac{1}{11}(A^2 - A - 9I) \dots \dots \dots (1)$$

Now

$$\begin{aligned} A^2 &= \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1+4+9 & 2-2+3 & 3+2+3 \\ 2-2+3 & 4+1+1 & 6-1+1 \\ 3+2+3 & 6-1+1 & 9+1+1 \end{pmatrix} \\ &= \begin{pmatrix} 14 & 3 & 8 \\ 3 & 6 & 6 \\ 8 & 6 & 11 \end{pmatrix} \end{aligned}$$

From (1) we have,

$$\begin{aligned} A^{-1} &= \frac{1}{11} \left(\begin{pmatrix} 14 & 3 & 8 \\ 3 & 6 & 6 \\ 8 & 6 & 11 \end{pmatrix} - \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix} - 9 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right) \\ &= \frac{1}{11} \left(\begin{pmatrix} 14 & 3 & 8 \\ 3 & 6 & 6 \\ 8 & 6 & 11 \end{pmatrix} - \begin{pmatrix} 1 & 2 & 3 \\ 2 & -1 & 1 \\ 3 & 1 & 1 \end{pmatrix} - \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix} \right) \\ &= \frac{1}{11} \begin{pmatrix} 4 & 1 & 5 \\ 1 & -2 & 5 \\ 5 & 5 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 4/11 & 1/11 & 5/11 \\ 1/11 & -2/11 & 5/11 \\ 5/11 & 5/11 & 1/11 \end{pmatrix} \end{aligned}$$

This is required inverse matrix of the given matrix. Similarly, we can find A^{-2} and A^{-3} . (Ans.)

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Exercise: Verify the Cayley-Hamilton Theorem for the following matrices and also find inverse matrices A^{-1} , A^{-2} & A^{-3} :

$$a). A = \begin{pmatrix} 2 & -2 & 1 \\ 2 & -8 & -2 \\ 1 & 2 & 2 \end{pmatrix} ; b). A = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix} ; c). A = \begin{pmatrix} 0 & 1 & 2 \\ 2 & 3 & 0 \\ 1 & 1 & -2 \end{pmatrix} .$$

Problem-03: Find the eigenvalues and associated eigenvectors of the matrix

$$A = \begin{pmatrix} -3 & 2 & 2 \\ -6 & 5 & 2 \\ -7 & 4 & 4 \end{pmatrix} .$$

Solution: The given matrix is,

$$A = \begin{pmatrix} -3 & 2 & 2 \\ -6 & 5 & 2 \\ -7 & 4 & 4 \end{pmatrix}$$

The characteristic matrix of A is,

$$\begin{aligned} A - \lambda I &= \begin{pmatrix} -3 & 2 & 2 \\ -6 & 5 & 2 \\ -7 & 4 & 4 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} -3 & 2 & 2 \\ -6 & 5 & 2 \\ -7 & 4 & 4 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} -3-\lambda & 2 & 2 \\ -6 & 5-\lambda & 2 \\ -7 & 4 & 4-\lambda \end{pmatrix} \end{aligned}$$

The characteristic polynomial of A is,

$$\begin{aligned} \Delta &= |A - \lambda I| \\ &= \begin{vmatrix} -3-\lambda & 2 & 2 \\ -6 & 5-\lambda & 2 \\ -7 & 4 & 4-\lambda \end{vmatrix} \\ &= (-3-\lambda)\{(5-\lambda)(4-\lambda)-8\} - 2\{-6(4-\lambda)+14\} + 2\{-24+7(5-\lambda)\} \\ &= (-3-\lambda)(20-9\lambda+\lambda^2-8) - 2(-24+6\lambda+14) + 2(-24+35-7\lambda) \\ &= (-3-\lambda)(\lambda^2-9\lambda+12) - 2(6\lambda-10) + 2(-7\lambda+11) \\ &= -3\lambda^2 + 27\lambda - 36 - \lambda^3 + 9\lambda^2 - 12\lambda - 12\lambda + 20 - 14\lambda + 22 \\ &= -\lambda^3 + 6\lambda^2 - 11\lambda + 6 \end{aligned}$$

The characteristic equation of A is,

$$|A - \lambda I| = 0$$

$$\Rightarrow -\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0$$

$$\begin{aligned}
 &\Rightarrow \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0 \\
 &\Rightarrow \lambda^3 - \lambda^2 - 5\lambda^2 + 5\lambda + 6\lambda - 6 = 0 \\
 &\Rightarrow \lambda^2(\lambda - 1) - 5\lambda(\lambda - 1) + 6(\lambda - 1) = 0 \\
 &\Rightarrow (\lambda - 1)(\lambda^2 - 5\lambda + 6) = 0 \\
 &\Rightarrow (\lambda - 1)(\lambda - 2)(\lambda - 3) = 0 \\
 &\therefore \lambda = 1, 2, 3
 \end{aligned}$$

The eigenvalues of $\mathcal{A}$ are 1, 2, 3.

2nd part:

Let $v_1 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ be a non-zero eigenvector corresponding to the eigenvalue $\lambda = 1$.

$$\begin{aligned}
 &\therefore (A - \lambda I)v_1 = 0 \\
 &\Rightarrow \begin{pmatrix} -4 & 2 & 2 \\ -6 & 4 & 2 \\ -7 & 4 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\
 &\quad \left. \begin{aligned} -4x + 2y + 2z &= 0 \\ \text{or, } -6x + 4y + 2z &= 0 \\ -7x + 4y + 3z &= 0 \end{aligned} \right\} \\
 &\quad \left. \begin{aligned} -2x + y + z &= 0 \\ \text{or, } -3x + 2y + z &= 0 \\ -7x + 4y + 3z &= 0 \end{aligned} \right\} \begin{aligned} L_1' &\rightarrow \frac{1}{2}L_1 \\ L_2' &\rightarrow \frac{1}{2}L_2 \end{aligned} \\
 &\quad \left. \begin{aligned} -2x + y + z &= 0 \\ \text{or, } y - z &= 0 \\ y - z &= 0 \end{aligned} \right\} \begin{aligned} L_2' &\rightarrow 2L_2 - 3L_1 \\ L_3' &\rightarrow 2L_3 - 7L_1 \end{aligned} \\
 &\quad \left. \begin{aligned} -2x + y + z &= 0 \\ \text{or, } y - z &= 0 \\ 0 &= 0 \end{aligned} \right\} L_3' \rightarrow L_3 - L_2 \\
 &\quad \left. \begin{aligned} -2x + y + z &= 0 \\ \text{or, } y - z &= 0 \end{aligned} \right\} L_3' \rightarrow L_3 - L_2
 \end{aligned}$$

There are 2 equations in 3 unknowns. So there is $(3-2) = 1$ free variable which is z . Thus the system has only one independent solution.

Putting $z = 1$ then we get $v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

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Thus the independent eigenvector is $v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ corresponding to the eigenvalue $\lambda_1 = 1$ and

$\{(1,1,1)\}$ is a basis of the eigenspace corresponding to the eigenvalue $\lambda_1 = 1$.

Again, Let $v_2 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ be a non-zero eigenvector corresponding to the eigenvalue $\lambda = 2$.

$$\therefore (A - \lambda I)v_2 = 0$$

$$\Rightarrow \begin{pmatrix} -5 & 2 & 2 \\ -6 & 3 & 2 \\ -7 & 4 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\left. \begin{array}{l} -5x + 2y + 2z = 0 \\ \text{or, } -6x + 3y + 2z = 0 \\ -7x + 4y + 2z = 0 \end{array} \right\}$$

$$\left. \begin{array}{l} -5x + 2y + 2z = 0 \\ \text{or, } \quad \quad 3y - 2z = 0 \\ \quad \quad \quad 6y - 4z = 0 \end{array} \right\} \begin{array}{l} L_2' \rightarrow 5L_2 - 6L_1 \\ L_3' \rightarrow 5L_3 - 7L_1 \end{array}$$

$$\left. \begin{array}{l} -5x + 2y + 2z = 0 \\ \text{or, } \quad \quad 3y - 2z = 0 \\ \quad \quad \quad 3y - 2z = 0 \end{array} \right\} L_3' \rightarrow \frac{1}{2}L_3$$

$$\left. \begin{array}{l} -5x + 2y + 2z = 0 \\ \text{or, } \quad \quad 3y - 2z = 0 \\ \quad \quad \quad 0 = 0 \end{array} \right\} L_3' \rightarrow L_3 - L_2$$

$$\left. \begin{array}{l} -5x + 2y + 2z = 0 \\ \text{or, } \quad \quad 3y - 2z = 0 \end{array} \right\}$$

There are 2 equations in 3 unknowns. So there is $(3-2) = 1$ free variable which is z . Thus the system has only one independent solution.

$$\text{Putting } z=3 \text{ then we get } v_2 = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}.$$

Thus the independent eigenvector is $v_2 = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}$ corresponding to the eigenvalue $\lambda_2 = 2$ and

$\{(2,2,3)\}$ is a basis of the Eigen space corresponding to the eigenvalue $\lambda_2 = 2$.

Again, let $v_3 = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ be a non-zero eigenvector corresponding to the eigenvalue $\lambda=3$.

$$\therefore (A - \lambda I)v_3 = 0$$

$$\Rightarrow \begin{pmatrix} -6 & 2 & 2 \\ -6 & 2 & 2 \\ -7 & 4 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\left. \begin{array}{l} -6x + 2y + 2z = 0 \\ \text{or, } -6x + 2y + 2z = 0 \\ -7x + 4y + z = 0 \end{array} \right\}$$

$$\text{or, } \left. \begin{array}{l} -6x + 2y + 2z = 0 \\ 0 = 0 \\ 10y - 8z = 0 \end{array} \right\} \begin{array}{l} L_2' \rightarrow L_2 - L_1 \\ L_3' \rightarrow 6L_3 - 7L_1 \end{array}$$

$$\text{or, } \left. \begin{array}{l} -6x + 2y + 2z = 0 \\ 10y - 8z = 0 \end{array} \right\}$$

$$\text{or, } \left. \begin{array}{l} -6x + 2y + 2z = 0 \\ 5y - 4z = 0 \end{array} \right\} L_2' \rightarrow \frac{1}{2}L_2$$

There are 2 equations in 3 unknowns. So there is $(3-2) = 1$ free variable which is z . Thus the system has only one independent solution.

$$\text{Putting } z=5 \text{ then we get } v_3 = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}.$$

Thus the independent eigenvector is $v_3 = \begin{pmatrix} 3 \\ 4 \\ 5 \end{pmatrix}$ corresponding to the eigenvalue $\lambda_3 = 3$

and $\{(3,4,5)\}$ is a basis of the Eigen space corresponding to the eigenvalue $\lambda_3 = 3$.

Problem-04: If a matrix is, $A = \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix}$

Then, (a). Find an spectrum of A .

(b). Find eigenvalues of A^T , A^3 , A^{-1} , A^{-3} .

(c). Find A^{-1} by using Cayley-Hamilton Theorem.

Solution: The given matrix is,

$$A = \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix}$$

The characteristic matrix of A is,

$$\begin{aligned} A - \lambda I &= \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} 1-\lambda & -3 & 3 \\ 3 & -5-\lambda & 3 \\ 6 & -6 & 4-\lambda \end{pmatrix} \end{aligned}$$

The characteristic polynomial of A is,

$$\begin{aligned} \Delta &= |A - \lambda I| \\ &= \begin{vmatrix} 1-\lambda & -3 & 3 \\ 3 & -5-\lambda & 3 \\ 6 & -6 & 4-\lambda \end{vmatrix} \\ &= (1-\lambda)\{(-5-\lambda)(4-\lambda)+18\} + 3\{3(4-\lambda)-18\} + 3\{-18-6(-5-\lambda)\} \\ &= (1-\lambda)(\lambda^2 + \lambda - 20 + 18) + 3(12 - 3\lambda - 18) + 3(-18 + 30 + 6\lambda) \\ &= (1-\lambda)(\lambda^2 + \lambda - 2) - 9\lambda - 18 + 36 + 18\lambda \\ &= \lambda^2 + \lambda - 2 - \lambda^3 - \lambda^2 + 2\lambda - 9\lambda - 18 + 36 + 18\lambda \\ &= -\lambda^3 + 12\lambda + 16 \end{aligned}$$

The characteristic equation of A is,

$$\begin{aligned} |A - \lambda I| &= 0 \\ \Rightarrow -\lambda^3 + 12\lambda + 16 &= 0 \\ \Rightarrow \lambda^3 - 12\lambda - 16 &= 0 \quad \dots \dots \dots (1) \\ \Rightarrow \lambda^3 + 2\lambda^2 - 2\lambda^2 - 4\lambda - 8\lambda - 16 &= 0 \\ \Rightarrow \lambda^2(\lambda + 2) - 2\lambda(\lambda + 2) - 8(\lambda + 2) &= 0 \\ \Rightarrow (\lambda + 2)(\lambda^2 - 2\lambda - 8) &= 0 \\ \Rightarrow (\lambda + 2)(\lambda + 2)(\lambda - 4) &= 0 \\ \therefore \lambda &= -2, -2, 4 \end{aligned}$$

The eigenvalues of A are -2, -2, 4.

(a). The spectrum is, $\{-2, 4\}$.

(b). The eigenvalues of the matrix A^T are, -2, -2, 4.

The eigenvalues of the matrix A^3 are, $(-2)^3, (-2)^3, 4^3$ or -8, -8, 64.

The eigenvalues of the matrix A^{-1} are, $(-2)^{-1}, (-2)^{-1}, 4^{-1}$ or $-\frac{1}{2}, -\frac{1}{2}, \frac{1}{4}$.

The eigenvalues of the matrix A^{-3} are, $(-2)^{-3}, (-2)^{-3}, 4^{-3}$ or $-\frac{1}{8}, -\frac{1}{8}, \frac{1}{64}$.

(c). From Eq. (1) we have,

$$\lambda^3 - 12\lambda - 16 = 0$$

By Cayley-Hamilton Theorem we have,

$$A^3 - 12A - 16I = 0$$

$$\Rightarrow A^2 - 12I - 16A^{-1} = 0$$

$$\Rightarrow -16A^{-1} = -A^2 + 12I$$

$$\Rightarrow A^{-1} = \frac{1}{16}A^2 - \frac{3}{4}I \quad \dots \dots \dots (2)$$

$$\begin{aligned} \text{Now, } A^2 &= \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix} \begin{pmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 1-9+18 & -3+15-16 & 3-9+12 \\ 3-15+18 & -9+25-18 & 9-15+12 \\ 6-18+24 & -18+30-24 & 18-18+16 \end{pmatrix} \\ &= \begin{pmatrix} 10 & -4 & 6 \\ 6 & -2 & 6 \\ 12 & -12 & 16 \end{pmatrix} \end{aligned}$$

From Eq. (2) we have,

$$\begin{aligned} A^{-1} &= \frac{1}{16} \begin{pmatrix} 10 & -4 & 6 \\ 6 & -2 & 6 \\ 12 & -12 & 16 \end{pmatrix} - \frac{3}{4} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} \frac{5}{8} & -\frac{1}{4} & \frac{3}{8} \\ \frac{3}{8} & -\frac{1}{8} & \frac{3}{8} \\ \frac{3}{4} & -\frac{3}{4} & 1 \end{pmatrix} - \begin{pmatrix} \frac{3}{4} & 0 & 0 \\ 0 & \frac{3}{4} & 0 \\ 0 & 0 & \frac{3}{4} \end{pmatrix} \\ &= \begin{pmatrix} -\frac{1}{8} & -\frac{1}{4} & \frac{3}{8} \\ \frac{3}{8} & -\frac{7}{8} & \frac{3}{8} \\ \frac{3}{4} & -\frac{3}{4} & \frac{1}{4} \end{pmatrix} \end{aligned}$$

Problem-05: If a matrix is, $A = \begin{pmatrix} 0 & 4 & 2 \\ -3 & 8 & 3 \\ 4 & -8 & -2 \end{pmatrix}$

Then, (a). Find an spectrum of A .

(b). Find eigenvalues of A^T, A^3, A^{-1}, A^{-3} .

(c). Find A^{-1} by using Cayley-Hamilton Theorem.

Solution: The given matrix is,

$$A = \begin{pmatrix} 0 & 4 & 2 \\ -3 & 8 & 3 \\ 4 & -8 & -2 \end{pmatrix}$$

The characteristic matrix of A is,

$$\begin{aligned} A - \lambda I &= \begin{pmatrix} 0 & 4 & 2 \\ -3 & 8 & 3 \\ 4 & -8 & -2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 4 & 2 \\ -3 & 8 & 3 \\ 4 & -8 & -2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} -\lambda & 4 & 2 \\ -3 & 8-\lambda & 3 \\ 4 & -8 & -2-\lambda \end{pmatrix} \end{aligned}$$

The characteristic polynomial of A is,

$$\begin{aligned} \Delta &= |A - \lambda I| \\ &= \begin{vmatrix} -\lambda & 4 & 2 \\ -3 & 8-\lambda & 3 \\ 4 & -8 & -2-\lambda \end{vmatrix} \\ &= -\lambda \{(8-\lambda)(-2-\lambda) + 24\} - 4\{-3(-2-\lambda) - 12\} + 2\{24 - 4(8-\lambda)\} \\ &= -\lambda(\lambda^2 - 6\lambda - 16 + 24) - 4(6 + 3\lambda - 12) + 2(24 - 32 + 4\lambda) \\ &= -\lambda(\lambda^2 - 6\lambda + 8) - 4(3\lambda - 6) + 2(4\lambda - 8) \\ &= -\lambda^3 + 6\lambda^2 - 8\lambda - 12\lambda + 24 + 8\lambda - 16 \\ &= -\lambda^3 + 6\lambda^2 - 12\lambda + 8 \end{aligned}$$

The characteristic equation of A is,

$$\begin{aligned} |A - \lambda I| &= 0 \\ \Rightarrow -\lambda^3 + 6\lambda^2 - 12\lambda + 8 &= 0 \\ \Rightarrow \lambda^3 - 6\lambda^2 + 12\lambda - 8 &= 0 \quad \dots \dots \dots (1) \\ \Rightarrow \lambda^3 - 2\lambda^2 - 4\lambda^2 + 8\lambda + 4\lambda - 8 &= 0 \end{aligned}$$

$$\Rightarrow \lambda^2(\lambda-2) - 4\lambda(\lambda-2) + 4(\lambda-2) = 0$$

$$\Rightarrow (\lambda-2)(\lambda^2 - 4\lambda + 4) = 0$$

$$\Rightarrow (\lambda-2)(\lambda-2)(\lambda-2) = 0$$

$$\therefore \lambda = 2, 2, 2$$

The eigenvalues of $\mathcal{A}$ are 2, 2, 2.

(a). The spectrum is, $\{2\}$.

(b). The eigenvalues of the matrix A^T are, 2, 2, 2.

The eigenvalues of the matrix A^3 are, $2^3, 2^3, 2^3$ or 8, 8, 8.

The eigenvalues of the matrix A^{-1} are, $2^{-1}, 2^{-1}, 2^{-1}$ or $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$.

The eigenvalues of the matrix A^{-3} are, $2^{-3}, 2^{-3}, 2^{-3}$ or $\frac{1}{8}, \frac{1}{8}, \frac{1}{8}$.

(c). From Eq. (1) we have,

$$\lambda^3 - 6\lambda^2 + 12\lambda - 8 = 0$$

By Cayley-Hamilton Theorem we have,

$$A^3 - 6A^2 + 12A - 8I = 0$$

$$\Rightarrow A^2 - 6A + 12I - 8A^{-1} = 0 \quad (\text{Multiplying by } A^{-1})$$

$$\Rightarrow -8A^{-1} = -A^2 + 6A - 12I$$

$$\Rightarrow A^{-1} = \frac{1}{8}A^2 - \frac{3}{4}A + \frac{3}{2}I \quad \dots \dots \dots (2)$$

$$\begin{aligned} \text{Now, } A^2 &= \begin{pmatrix} 0 & 4 & 2 \\ -3 & 8 & 3 \\ 4 & -8 & -2 \end{pmatrix} \begin{pmatrix} 0 & 4 & 2 \\ -3 & 8 & 3 \\ 4 & -8 & -2 \end{pmatrix} \\ &= \begin{pmatrix} 0-12+8 & 0+32-16 & 0+12-4 \\ 0-24+12 & -12+64-24 & -6+24-6 \\ 0+24-8 & 16-64+16 & 8-24+4 \end{pmatrix} \\ &= \begin{pmatrix} -4 & 16 & 8 \\ -12 & 28 & 12 \\ 16 & -32 & -12 \end{pmatrix} \end{aligned}$$

From Eq. (2) we have,

$$A^{-1} = \frac{1}{8} \begin{pmatrix} -4 & 16 & 8 \\ -12 & 28 & 12 \\ 16 & -32 & -12 \end{pmatrix} - \frac{3}{4} \begin{pmatrix} 0 & 4 & 2 \\ -3 & 8 & 3 \\ 4 & -8 & -2 \end{pmatrix} + \frac{3}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{1}{2} & 2 & 1 \\ -\frac{3}{2} & \frac{7}{2} & \frac{3}{2} \\ 2 & -4 & -\frac{3}{2} \end{pmatrix} - \begin{pmatrix} 0 & 3 & \frac{3}{2} \\ -\frac{9}{4} & 6 & \frac{9}{4} \\ 3 & -6 & -\frac{3}{2} \end{pmatrix} + \begin{pmatrix} \frac{3}{2} & 0 & 0 \\ 0 & \frac{3}{2} & 0 \\ 0 & 0 & \frac{3}{2} \end{pmatrix}$$

$$= \begin{pmatrix} 1 & -1 & -\frac{1}{2} \\ \frac{3}{4} & 2 & -\frac{3}{4} \\ -1 & 2 & \frac{3}{2} \end{pmatrix}$$

Problem-o6: If $A = \begin{pmatrix} 1 & 2 & 2 \\ 1 & 2 & -1 \\ -1 & 1 & 4 \end{pmatrix}$, then find its $\begin{cases} \text{i). eigenvalues} \\ \text{ii). algebraic multiplicities of eigenvalues} \\ \text{iii). geometric multiplicities.} \end{cases}$

Solution: The given matrix is,

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 1 & 2 & -1 \\ -1 & 1 & 4 \end{pmatrix}$$

The characteristic matrix of A is,

$$A - \lambda I = \begin{pmatrix} 1 & 2 & 2 \\ 1 & 2 & -1 \\ -1 & 1 & 4 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 2 \\ 1 & 2 & -1 \\ -1 & 1 & 4 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix}$$

$$= \begin{pmatrix} 1-\lambda & 2 & 2 \\ 3 & 2-\lambda & -1 \\ -1 & 1 & 4-\lambda \end{pmatrix}$$

The characteristic polynomial of A is,

$$\Delta = |A - \lambda I|$$

$$= \begin{vmatrix} 1-\lambda & 2 & 2 \\ 3 & 2-\lambda & -1 \\ -1 & 1 & 4-\lambda \end{vmatrix}$$

$$= (1-\lambda)\{(2-\lambda)(4-\lambda)+1\} - 2\{(4-\lambda)-1\} + 2\{1+(2-\lambda)\}$$

$$= (1-\lambda)(8-6\lambda+\lambda^2+1) - 2(3-\lambda) + 2(3-\lambda) = (1-\lambda)(9-6\lambda+\lambda^2) - 6 + 2\lambda + 6 - 2\lambda$$

$$= (1-\lambda)(\lambda^2 - 6\lambda + 9)$$

$$= (1-\lambda)(\lambda-3)(\lambda-3)$$

The characteristic equation of A is,

Eigen Values and Eigen Vectors of Matrices

$$\begin{aligned} |A - \lambda I| &= 0 \\ \Rightarrow (1 - \lambda)(\lambda^2 - 6\lambda + 9) &= 0 \\ \Rightarrow (1 - \lambda)(\lambda - 3)(\lambda - 3) &= 0 \\ \therefore \lambda &= 1, 3, 3 \end{aligned}$$

The eigenvalues of $\mathcal{A}$ are 1, 3, 3.

The algebraic multiplicity of $\lambda=1$ is 1. So its geometric multiplicity must be 1.

The algebraic multiplicity of $\lambda=3$ is 2. So its geometric multiplicity may be 1 or, 2.

To become sure we use $v = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ in the characteristic equation.

$$\begin{aligned} \text{i.e., } (A - \lambda I)v &= 0 \\ \Rightarrow \begin{pmatrix} -2 & 2 & 2 \\ 1 & -1 & -1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} &= 0 \end{aligned}$$

$$\text{or, } \begin{cases} -2a + 2b + 2c = 0 \\ a - b - c = 0 \\ -a + b + c = 0 \end{cases}$$

$$\text{or, } \begin{cases} -2a + 2b + 2c = 0 \\ 0 = 0 \\ 0 = 0 \end{cases} \begin{cases} L_2' = 2L_2 + L_1 \\ L_3' = 2L_3 - L_1 \end{cases}$$

$$\text{or, } -2a + 2b + 2c = 0$$

$$\text{or, } a - b - c = 0$$

$$\text{or, } a = b + c$$

$$\therefore v = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$\Rightarrow v = \begin{pmatrix} b+c \\ b \\ c \end{pmatrix}$$

$$\Rightarrow v = b \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \text{ where, } b \neq 0 \text{ and } c \neq 0$$

Here, the resulting eigenspace is 2-dimensional.

Hence, the geometric multiplicity of $\lambda=3$ is 2.

Problem-07: Find the eigenvalues of the matrix $A = \begin{pmatrix} -3 & 1 & -1 \\ -7 & 5 & -1 \\ -6 & 6 & -2 \end{pmatrix}$.

Solution: The given matrix is,

$$A = \begin{pmatrix} -3 & 1 & -1 \\ -7 & 5 & -1 \\ -6 & 6 & -2 \end{pmatrix}$$

The characteristic matrix of A is,

$$\begin{aligned} A - \lambda I &= \begin{pmatrix} -3 & 1 & -1 \\ -7 & 5 & -1 \\ -6 & 6 & -2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} -3 & 1 & -1 \\ -7 & 5 & -1 \\ -6 & 6 & -2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} -3-\lambda & 1 & -1 \\ -7 & 5-\lambda & -1 \\ -6 & 6 & -2-\lambda \end{pmatrix} \end{aligned}$$

The characteristic polynomial of A is,

$$\begin{aligned} \Delta &= |A - \lambda I| \\ &= \begin{vmatrix} -3-\lambda & 1 & -1 \\ -7 & 5-\lambda & -1 \\ -6 & 6 & -2-\lambda \end{vmatrix} \\ &= (-3-\lambda)\{(5-\lambda)(-2-\lambda)+6\} - \{-7(-2-\lambda)-6\} - \{-42+6(5-\lambda)\} \\ &= (-3-\lambda)\{\lambda^2-3\lambda-10\}+6 - (14+7\lambda-6) - (-42+30-6\lambda) \\ &= (-3-\lambda)(\lambda^2-3\lambda-4) - (7\lambda+8) - (-12-6\lambda) \\ &= -3\lambda^2+9\lambda+12-\lambda^3+3\lambda^2+4\lambda-7\lambda-8+12+6\lambda \\ &= -\lambda^3+12\lambda+16 \end{aligned}$$

The characteristic equation of A is,

$$\begin{aligned} |A - \lambda I| &= 0 \\ \Rightarrow -\lambda^3 + 12\lambda + 16 &= 0 \\ \Rightarrow \lambda^3 - 12\lambda - 16 &= 0 \\ \Rightarrow \lambda^3 - 4\lambda^2 + 4\lambda^2 - 16\lambda + 4\lambda - 16 &= 0 \\ \Rightarrow \lambda^2(\lambda-4) + 4\lambda(\lambda-4) + 4(\lambda-4) &= 0 \\ \Rightarrow (\lambda-4)(\lambda^2+4\lambda+4) &= 0 \\ \Rightarrow (\lambda-4)(\lambda+2)(\lambda+2) &= 0 \end{aligned}$$

$$\therefore \lambda = -2, -2, 4$$

The eigenvalues of $\mathcal{A}$ are -2, -2, 4.

Problem-o8: If a matrix is, $A = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & 4 \end{pmatrix}$.

Then, (a). Find an spectrum of A .

(b). Find eigenvalues of A^T , A^3 , A^{-1} , A^{-3} .

(c). Find A^{-1} by using Cayley-Hamilton Theorem.

Solution: The given matrix is,

$$A = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & 4 \end{pmatrix}$$

The characteristic matrix of $\mathcal{A}$ is,

$$\begin{aligned} A - \lambda I &= \begin{pmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & 4 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & 4 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} 1-\lambda & 0 & -2 \\ 0 & -\lambda & 0 \\ -2 & 0 & 4-\lambda \end{pmatrix} \end{aligned}$$

The characteristic polynomial of $\mathcal{A}$ is,

$$\begin{aligned} \Delta &= |A - \lambda I| \\ &= \begin{vmatrix} 1-\lambda & 0 & -2 \\ 0 & -\lambda & 0 \\ -2 & 0 & 4-\lambda \end{vmatrix} \\ &= (1-\lambda) \{ -\lambda(4-\lambda) - 0 \} - 0 - 2(0 - 2\lambda) \\ &= (1-\lambda)(-4\lambda + \lambda^2) + 4\lambda \\ &= -4\lambda + \lambda^2 + 4\lambda^2 - \lambda^3 + 4\lambda \\ &= -\lambda^3 + 5\lambda^2 \end{aligned}$$

The characteristic equation of $\mathcal{A}$ is,

$$\begin{aligned} |A - \lambda I| &= 0 \\ \Rightarrow -\lambda^3 + 5\lambda^2 &= 0 \\ \Rightarrow \lambda^3 - 5\lambda^2 &= 0 \\ \Rightarrow \lambda^2(\lambda - 5) &= 0 \end{aligned}$$

$$\therefore \lambda = 0, 0, 5$$

The eigenvalues of $\mathcal{A}$ are 0, 0, 5.

(a). The spectrum is, $\{0, 5\}$.

(b). The eigenvalues of the matrix A^T are, 0, 0, 5.

The eigenvalues of the matrix A^3 are, $0^3, 0^3, 5^3$ or 0, 0, 125.

Since, 0 is the eigenvalue of the matrix $\mathcal{A}$ so it is a singular matrix and we know that singular matrix does not possess inverse matrix. So, the eigenvalues of matrices A^{-1} & A^{-3} are not possible.

(c). Since, the matrix is singular so its inverse does not exist.

Problem-09: Find the eigenvalues of the matrix $A = \begin{pmatrix} 2 & 1 & 0 \\ 3 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}$.

Then, (a). Find an spectrum of A .

(b). Find eigenvalues of A^T , A^3 , A^{-1} , A^{-3} .

(c). Find A^{-1} by using Cayley-Hamilton Theorem.

Solution: The given matrix is,

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 3 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

The characteristic matrix of $\mathcal{A}$ is,

$$\begin{aligned} A - \lambda I &= \begin{pmatrix} 2 & 1 & 0 \\ 3 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 1 & 0 \\ 3 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} 2-\lambda & 1 & 0 \\ 3 & 2-\lambda & 0 \\ 0 & 0 & 4-\lambda \end{pmatrix} \end{aligned}$$

The characteristic polynomial of $\mathcal{A}$ is,

$$\begin{aligned} \Delta &= |A - \lambda I| \\ &= \begin{vmatrix} 2-\lambda & 1 & 0 \\ 3 & 2-\lambda & 0 \\ 0 & 0 & 4-\lambda \end{vmatrix} \\ &= (2-\lambda)\{(2-\lambda)(4-\lambda)-0\} - \{3(4-\lambda)-0\} + 0 \\ &= (2-\lambda)^2(4-\lambda) - 3(4-\lambda) \end{aligned}$$

$$\begin{aligned}
 &= (4-\lambda)\{(2-\lambda)^2-3\} \\
 &= (4-\lambda)(4-4\lambda+\lambda^2-3) \\
 &= (4-\lambda)(\lambda^2-4\lambda+1)
 \end{aligned}$$

The characteristic equation of A is,

$$|A-\lambda I|=0$$

$$\Rightarrow (4-\lambda)(\lambda^2-4\lambda+1)=0$$

$$\therefore 4-\lambda=0 \text{ or, } \lambda^2-4\lambda+1=0$$

$$\Rightarrow \lambda=4 \text{ or, } \lambda=\frac{4\pm\sqrt{16-4}}{2}$$

$$\text{or, } \lambda=\frac{4\pm\sqrt{12}}{2}$$

$$\text{or, } \lambda=\frac{4\pm2\sqrt{3}}{2}$$

$$\text{or, } \lambda=2\pm\sqrt{3}$$

$$\therefore \lambda=4, 2+\sqrt{3}, 2-\sqrt{3}$$

The eigenvalues of A are $4, 2+\sqrt{3}, 2-\sqrt{3}$.

(a). The spectrum is, $\{4, 2+\sqrt{3}, 2-\sqrt{3}\}$.

(b). The eigenvalues of the matrix A^T are, $4, 2+\sqrt{3}, 2-\sqrt{3}$.

The eigenvalues of the matrix A^3 are, $4^3, (2+\sqrt{3})^3, (2-\sqrt{3})^3$ or $64, (2+\sqrt{3})^3, (2-\sqrt{3})^3$.

The eigenvalues of the matrix A^{-1} are, $4^{-1}, (2+\sqrt{3})^{-1}, (2-\sqrt{3})^{-1}$.

The eigenvalues of the matrix A^{-3} are, $4^{-3}, (2+\sqrt{3})^{-3}, (2-\sqrt{3})^{-3}$.

(c). Try yourself.

Problem-10: Find the eigenvalues of the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -2 & 1 \end{pmatrix}$.

Solution: The given matrix is,

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -2 & 1 \end{pmatrix}$$

The characteristic matrix of A is,

$$\begin{aligned}
 A - \lambda I &= \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -2 & 1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & -2 & 1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \\
 &= \begin{pmatrix} 1-\lambda & 2 & 3 \\ 0 & 1-\lambda & 2 \\ 0 & -2 & 1-\lambda \end{pmatrix}
 \end{aligned}$$

The characteristic polynomial of $\mathcal{A}$ is,

$$\begin{aligned}
 \Delta &= |A - \lambda I| \\
 &= \begin{vmatrix} 1-\lambda & 2 & 3 \\ 0 & 1-\lambda & 2 \\ 0 & -2 & 1-\lambda \end{vmatrix} \\
 &= (1-\lambda) \{ (1-\lambda)^2 + 4 \} \\
 &= (1-\lambda) (\lambda^2 - 2\lambda + 1 + 4) \\
 &= (1-\lambda) (\lambda^2 - 2\lambda + 5)
 \end{aligned}$$

The characteristic equation of $\mathcal{A}$ is,

$$\begin{aligned}
 |A - \lambda I| &= 0 \\
 \Rightarrow (1-\lambda) (\lambda^2 - 2\lambda + 5) &= 0 \\
 \therefore 1-\lambda = 0 \quad \text{or, } \lambda^2 - 2\lambda + 5 &= 0 \\
 \Rightarrow \lambda = 1 \quad \text{or, } \lambda &= \frac{2 \pm \sqrt{4-20}}{2} \\
 \text{or, } \lambda &= \frac{2 \pm \sqrt{-16}}{2} \\
 \text{or, } \lambda &= \frac{2 \pm \sqrt{16i^2}}{2} \\
 \text{or, } \lambda &= \frac{2 \pm 4i}{2} \\
 \text{or, } \lambda &= 1 \pm 2i \\
 \therefore \lambda &= 1, 1 \pm 2i
 \end{aligned}$$

The eigenvalues of $\mathcal{A}$ are $1, 1 \pm 2i$.

Problem-11: Find the eigenvalues of the matrix $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix}$.

Solution: The given matrix is,

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix}$$

The characteristic matrix of A is,

$$\begin{aligned} A - \lambda I &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & -1 \\ 0 & 1 & 1-\lambda \end{pmatrix} \end{aligned}$$

The characteristic polynomial of A is,

$$\begin{aligned} \Delta &= |A - \lambda I| \\ &= \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & -1 \\ 0 & 1 & 1-\lambda \end{vmatrix} \\ &= (1-\lambda) \{ (1-\lambda)^2 + 1 \} \\ &= (1-\lambda) (\lambda^2 - 2\lambda + 2) \end{aligned}$$

The characteristic equation of A is,

$$\begin{aligned} |A - \lambda I| &= 0 \\ \Rightarrow (1-\lambda) (\lambda^2 - 2\lambda + 2) &= 0 \\ \therefore 1-\lambda &= 0 \quad \text{or} \quad \lambda^2 - 2\lambda + 2 = 0 \\ \Rightarrow \lambda &= 1 \quad \text{or} \quad \lambda = \frac{2 \pm \sqrt{4-8}}{2} \\ \text{or, } \lambda &= \frac{2 \pm \sqrt{-4}}{2} \\ \text{or, } \lambda &= \frac{2 \pm \sqrt{4i^2}}{2} \\ \text{or, } \lambda &= \frac{2 \pm 2i}{2} \\ \text{or, } \lambda &= 1 \pm i \\ \therefore \lambda &= 1, 1 \pm i \end{aligned}$$

The eigenvalues of A are $1, 1 \pm i$.

Problem-12: Find the eigenvalues of the matrix $A = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -2 & 3 & 0 & 0 \\ 2 & 0 & 4 & 2 \\ 1 & 3 & -2 & -1 \end{pmatrix}$

Solution: The given matrix is,

$$A = \begin{pmatrix} 2 & -1 & 0 & 0 \\ -2 & 3 & 0 & 0 \\ 2 & 0 & 4 & 2 \\ 1 & 3 & -2 & -1 \end{pmatrix}$$

The characteristic matrix of A is,

$$\begin{aligned} A - \lambda I &= \begin{pmatrix} 2 & -1 & 0 & 0 \\ -2 & 3 & 0 & 0 \\ 2 & 0 & 4 & 2 \\ 1 & 3 & -2 & -1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 2 & -1 & 0 & 0 \\ -2 & 3 & 0 & 0 \\ 2 & 0 & 4 & 2 \\ 1 & 3 & -2 & -1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 & 0 \\ 0 & \lambda & 0 & 0 \\ 0 & 0 & \lambda & 0 \\ 0 & 0 & 0 & \lambda \end{pmatrix} \\ &= \begin{pmatrix} 2-\lambda & -1 & 0 & 0 \\ -2 & 3-\lambda & 0 & 0 \\ 2 & 0 & 4-\lambda & 2 \\ 1 & 3 & -2 & -1-\lambda \end{pmatrix} \end{aligned}$$

The characteristic polynomial of A is,

$$\begin{aligned} \Delta &= |A - \lambda I| \\ &= \begin{vmatrix} 2-\lambda & -1 & 0 & 0 \\ -2 & 3-\lambda & 0 & 0 \\ 2 & 0 & 4-\lambda & 2 \\ 1 & 3 & -2 & -1-\lambda \end{vmatrix} \\ &= (2-\lambda) \begin{vmatrix} 3-\lambda & 0 & 0 \\ 0 & 4-\lambda & 2 \\ 3 & -2 & -1-\lambda \end{vmatrix} + 1 \begin{vmatrix} -2 & 0 & 0 \\ 2 & 4-\lambda & 2 \\ 1 & -2 & -1-\lambda \end{vmatrix} \\ &= (2-\lambda)(3-\lambda) \begin{vmatrix} 4-\lambda & 2 \\ -2 & -1-\lambda \end{vmatrix} - 2 \begin{vmatrix} 4-\lambda & 2 \\ -2 & -1-\lambda \end{vmatrix} \\ &= (2-\lambda)(3-\lambda) \{ (4-\lambda)(-1-\lambda) + 4 \} - 2 \{ (4-\lambda)(-1-\lambda) + 4 \} \\ &= (2-\lambda)(3-\lambda) (\lambda^2 - 3\lambda - 4 + 4) - 2 (\lambda^2 - 3\lambda - 4 + 4) \\ &= (2-\lambda)(3-\lambda) (\lambda^2 - 3\lambda) - 2 (\lambda^2 - 3\lambda) \end{aligned}$$

$$\begin{aligned}
 &= (\lambda^2 - 3\lambda)\{(2-\lambda)(3-\lambda)-2\} \\
 &= (\lambda^2 - 3\lambda)(\lambda^2 - 5\lambda + 6 - 2) \\
 &= (\lambda^2 - 3\lambda)(\lambda^2 - 5\lambda + 4) \\
 &= \lambda(\lambda-1)(\lambda-3)(\lambda-4)
 \end{aligned}$$

The characteristic equation of $\mathcal{A}$ is,

$$\begin{aligned}
 |A - \lambda I| &= 0 \\
 \Rightarrow \lambda(\lambda-1)(\lambda-3)(\lambda-4) &= 0 \\
 \therefore \lambda = 0; \lambda-1=0; \lambda-3=0; \lambda-4=0 \\
 \Rightarrow \lambda = 0; \lambda=1; \lambda=3; \lambda=4 \\
 \therefore \lambda = 0, 1, 3, 4
 \end{aligned}$$

The eigenvalues of $\mathcal{A}$ are 0, 1, 3, 4. (Ans)

Exercise: Find an spectrum & eigenvalues of A^T , A^3 , A^{-1} , A^{-3} . Also find A^{-1} by using Cayley-Hamilton Theorem. Where,

$$\begin{aligned}
 a). A &= \begin{pmatrix} 2 & -2 & 1 \\ 2 & -8 & -2 \\ 1 & 2 & 2 \end{pmatrix} & b). A &= \begin{pmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{pmatrix} & c). A &= \begin{pmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{pmatrix} & d). A &= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 4 & -17 & 8 \end{pmatrix} \\
 e). A &= \begin{pmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 3 \end{pmatrix} & f). A &= \begin{pmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{pmatrix} & g). A &= \begin{pmatrix} 1 & 2 & -1 \\ 0 & -2 & 0 \\ 0 & -5 & 2 \end{pmatrix} & h). A &= \begin{pmatrix} 5 & 0 & 0 \\ -1 & 3 & 0 \\ 0 & 2 & -1 \end{pmatrix} \\
 i). A &= \begin{pmatrix} 1 & 2 & 3 \\ 0 & -4 & 2 \\ 0 & 0 & 7 \end{pmatrix} & j). A &= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} & k). A &= \begin{pmatrix} 1 & 2 & 2 \\ 1 & 2 & -1 \\ -1 & 1 & 4 \end{pmatrix} & l). A &= \begin{pmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{pmatrix}
 \end{aligned}$$